

Each question is worth 5 points for a total of 35 points. The extra Credit can get you an additional 5 points making the highest possible score 40 out of 35. Write solutions using COMPLETE sentences.

Problem 1

Let A and B be sets, and let $f : A \rightarrow B$. Prove $f \circ id_A = id_B \circ f = f$.

Problem 2

Let A and B be sets and suppose $f : A \rightarrow B$ is one-to-one and onto. Prove $f \circ f^{-1} = id_B$ and $f^{-1} \circ f = id_A$.

Problem 3

A pair of fair dice are rolled.

- a) Write down the event space (S, P) .
- b) Write down the event A , that the sum of the dice is 7.
- c) Write down the event B , that neither die shows a 2.
- d) Evaluate $P(A)$
- e) Evaluate $P(B)$.
- f) What is the probability that they sum to 7 given that neither die shows a 2?

Problem 4

A fair coin is flipped three times. This is modeled by a sample space (S, P) where S contains the eight lists from HHH to TTT , all with probability $\frac{1}{8}$. Let X denote the number of times we see *TAILS*.

- a) Write X explicitly as a function defined on S .
- b) Write the event X is odd as a set.
- c) Calculate $P(X \text{ is odd})$.

Problem 5

Suppose a , b , and $p \in \mathbb{Z}$ where p is a prime. Prove that if $p|ab$, then $p|a$ or $p|b$.

Problem 6

Suppose $p, q_1, q_2, \dots, q_t$ are prime numbers. Prove that if $p|(q_1 \cdot q_2 \cdot \dots \cdot q_t)$ then $p = q_i$ for $1 \leq i \leq t$.

Problem 7

Let (S, P) be the sample space in which $S = \{1, 2, 3, 4\}$ and, for $k = 1, 2, 3, 4$, we have $P(k) = \frac{c}{k^2}$ where c is some number. Find c .

Extra Credit:

Let $f : \mathbb{Z} \rightarrow \mathbb{Z}$ by $f(x) = x^3$. Is f one-to-one? Is f onto? Prove your answers.